Hamid Dashti

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EDUCATION

Ph.D. in Geosciences, Boise State University (Idaho, USA) 2014 – 2019

M.Sc. in Remote Sensing and GIS, Shahid Beheshti University (Iran) 2008 – 2011

B.Sc. in Natural Resources Engineering, Gorgan University (Iran) 2003 – 2008

Experience

University of Chicago 2025 – Present

Computational Scientist, Research Computing Center

- Collaborating with research groups across campus on environmental sciences, remote sensing, and GeoAI projects.
- Porting and running CESM2 (atmosphere and land components) on Midway HPC for nudging experiments and performance optimization.
- Analyzing drone-based optical and LiDAR imagery of Colombian forests for anthropological and ecological research.
- Designing and leading workshops and hackathons on advanced computational methods, Python, R and research reproducibility.
- Advancing GeoAI research through collaborations with the UChicago Spatial Data Science and Data Science Institutes.

University of Wisconsin-Madison 2021 – 2025

Postdoctoral Research Associate, Department of Forest and Wildlife Ecology

- **Lead developer and writer on a NASA ROSES proposal ($800K, funded);** unable to remain formally listed due to transition to UChicago during submission.
- Secured a **$150K Postdoctoral fellowship (Co-I)** to integrate GCAM with Earth system models and assess carbon mitigation impacts on the dairy industry.
- Synthesized data from 300+ eddy covariance sites with satellite observations, revealing a non-linear relationship between biome structure and maximum light-use efficiency (Dashti et al, EL).
- Initiated a UW-Madison and Chicago Botanic Garden partnership for large-scale data collection and public outreach.
- Mentored PhD students in research on global change ecology, geospatial data science, and integrated assessment modeling, leading to multiple publications to date.
- Led a workshop on satellite and climate data processing using python in UW-Madison Research Bazaar 2024 and member of planning committee for Research Bazaar 2025.

University of Arizona 2019 – 2021

Postdoctoral Research Associate, School of Natural Resources and the Environment

- Led a 30-year Landsat analysis of Arctic–Boreal disturbance and recovery, introducing the concept of counterfactual recovery (Dashti et al. GRL; Dashti et al. ERL).
- Assimilated Earth observation data into CLM5, improving high-latitude carbon cycle simulation.
- Managed remote sensing instrumentation and measurments (spectrometers, cameras, SfM) for the **RainMann rainfall manipulation experiment**.
- Contributed to NASA's Surface Biology and Geology mission, coauthoring a widely cited algorithms perspective.
- Chaired the UA Postdoctoral Association outreach program, launching initiatives for high school students and co-organizing the Arizona Postdoc Research Conference.

Spatial Sciences Innovators 2012 – 2014

Co-founder, Science Team Leader, Tehran, Iran

- **Co-founded** a geospatial analytics startup and led the science team, proposal development, and client engagement.
- Developed a satellite-based forest monitoring system adopted nationally by Iran's Natural Resources and Watershed Management Organization.
- Directed projects on geospatial site selection, environmental impact assessment, and remote sensing applications.
- Designed and delivered educational workshops on geospatial data for natural resources, ranging from introductory to advanced levels.

PUBLICATIONS

[1] Yang, X., Guan, K., Chen, M., Johnson, J. E., Li, R., Kimm, H., Wu, G., Kim, J., **Dashti, H**., Wang, S., Lerdau, M., Frankenberg, C., & Berry, J. A. (2026). Sensing plant photosynthesis using solar-induced chlorophyll fluorescence: From chloroplasts to the globe. **Annual Review of Plant Biology**, 77 (in press).

[2] Ji, F., Zheng, T., Shiklomanov, A. N., Yang, R., Townsend, P. A., Li, F., Hao, D., **Dashti, H**., Kovach, K. R., You, H., Zhou, J., & Chen, M. (2026). Tracking seasonal variability in plant traits from spaceborne PRISMA and NEON AOP across forest types and ecoregions. **Remote Sensing of Environment**, 333, 115149. https://doi.org/10.1016/j.rse.2025.115149

[3] Liu, H., Pierrat, Z. A., **Dashti, H.**, & Chen, M. (2025). Unrevealing site-dependent relationship between solar-induced chlorophyll fluorescence and gross primary productivity using the terrestrial ecosystem carbon cycle simulator. **Remote Sensing of Environment, 331**, 115052. https://doi.org/10.1016/j.rse.2025.115052

[4] **Dashti, H**., Luo, L., Nicholson, C., Chen, M. (2025). Climate-driven ecosystem productivity changes restructure food systems. **Preprint, bioRxiv**, October 15, 2025. https://doi.org/10.1101/2025.10.14.682485 (Submitted to One Earth)

[5] **Dashti, H.**, Chen, M., Hao, D., Yang, X. (2025). Canopy structure exhibits linear and nonlinear links to biome-level maximum light use efficiency. **Ecology Letters**, 28, e70142. https://doi.org/10.1111/ele.70142

[6] Liu, H., Li, F., **Dashti, H.**, Chen, M. (2025). Hyperspectral surface reflectance improves GPP estimation in terrestrial biosphere modeling using model–data fusion. **Remote Sensing of Environment**, 330, 114989. https://doi.org/10.1016/j.rse.2025.114989

[7] Luo, M., Zhao, X., Hao, D., Bond-Lamberty, B., Daigneault, A., Patel, P.L., Kou-Giesbrecht, S., Reyer, C.P.O., **Dashti, H.**, Chen, M. (2025). Role of forest carbon change in shaping future land use and land cover change. **Global Change Biology**, 31, e70219. https://doi.org/10.1111/gcb.70219

[8] Ji, F., Li, F., **Dashti, H.**, Hao, D., Townsend, P.A., Zheng, T., You, H., Chen, M. (2025). Leveraging transfer learning and leaf spectroscopy for leaf trait prediction with broad spatial, species, and temporal applicability. ***Remote Sensing of Environment***, 326, 114818. https://doi.org/10.1016/j.rse.2025.114818

[9] **Dashti**, H., Chen, M., Smith, W.K., Zhao, K., Moore, D.J.P. (2024). Ecosystems Disturbance Recovery: What It Was or What It Could Have Been? **Geophysical Research Letters**, 51, e2024GL109219. https://doi.org/10.1029/2024GL109219

[10] Huo, X., Fox, A.M., **Dashti**, H., Devine, C., Gallery, W., Smith, W.K., et al. (2024). Integrating state data assimilation and innovative model parameterization reduces simulated carbon uptake in the Arctic and Boreal region. Journal of Geophysical Research: **Biogeosciences**, 129, e2024JG008004. https://doi.org/10.1029/2024JG008004

[11] Ji, F., Li, F., Hao, D., Shiklomanov, A.N., Yang, X., Townsend, P.A., **Dashti**, H., Nakaji, T., Kovach, K.R., Liu, H., Luo, M., Chen, M. (2024). Unveiling the transferability of PLSR models for leaf trait estimation: lessons from a comprehensive analysis with a novel global dataset. **New Phytologist**, 243, 111-131. https://doi.org/10.1111/nph.19807

[12] Enterkine, J., **Dashti**, H., Caughlin, T.T., Glenn, N.F. (2023). Applied soft classes and fuzzy confusion in a patchwork semi-arid ecosystem: stitching together classification techniques to preserve ecologically-meaningful information. **Remote Sensing of Environment**.

[13] Luo, M., Li, F., Hao, D., Zhu, Q., **Dashti**, H., Chen, M. (2023). Uncertain spatial pattern of future land use and land cover change and its impacts on terrestrial carbon cycle over the Arctic–Boreal region of North America. **Earth's Future**, 11, e2023EF003648. https://doi.org/10.1029/2023EF00364

[14] **Dashti**, H., Smith, W.K., Huo, X., Fox, A., Javadian, M., Devine, C.J., Behrangi, A., Moore, D.J.P. (2022). Underestimation of the impact of land cover change on the biophysical environment of the Arctic and Boreal Region of North America. **Environmental Research Letters**. http://iopscience.iop.org/article/10.1088/1748-9326/ac8da7

[15] 7. Fox, A.M., Huo, X., Hoar, T.J., **Dashti**, H., Smith, W.K., MacBean, N., et al. (2022). Assimilation of global satellite leaf area estimates reduces modeled global carbon uptake and energy loss by terrestrial ecosystems. Journal of Geophysical Research: **Biogeosciences**, 127, e2022JG006830. https://doi.org/10.1029/2022JG006830

[16] Ilangakoon, N., Glenn, N.F., Schneider, F.D., **Dashti**, H., Hancock, S., Spaete, L., Goulden, T. (2021). Airborne and Spaceborne Lidar Reveal Trends and Patterns of Functional Diversity in a Semi-Arid Ecosystem. **Frontiers in Remote Sensing**, 2, 743320. https://doi.org/10.3389/frsen.2021.743320

[17] Cawse-Nicholson, K., Townsend, P.A., Schimel, D., Assiri, A.M., Blake, P.L., Buongiorno, M.F., et al. (2021). NASA's Surface Biology and Geology Designated Observable: A Perspective on Surface Imaging Algorithms. **Remote Sensing of Environment**, 257, 112349. https://doi.org/10.1016/j.rse.2021.112349

[18] Pandit, K., **Dashti**, H., Hudak, A.T., Glenn, N.F., Flores, A.N., Shinneman, D.J. (2021). Understanding the effect of fire on vegetation composition and gross primary production in a semi-arid shrubland ecosystem using the Ecosystem Demography (EDv2.2) model. **Biogeosciences** Discussions, 2021, 1-20.

[19] **Dashti**, H., Pandit, K., Glenn, N.F., Shinneman, D.J., Flerchinger, G.N., Hudak, A.T., de Graaf, M.A., Flores, A., Ustin, S., Ilangakoon, N., Fellows, A.W. (2021). Performance of the ecosystem demography model (EDv2.2) in simulating gross primary production capacity and activity in a dryland study area. **Agricultural and Forest Meteorology**, 297, 108270.

[20] **Dashti**, H., Glenn, N.F., Ustin, S., Mitchell, J.J., Qi, Y., Ilangakoon, N.T., Flores, A.N., Silván-Cárdenas, J.L., Zhao, K., Spaete, L.P., de Graaff, M. (2019). Empirical Methods for Remote Sensing of Nitrogen in Drylands May Lead to Unreliable Interpretation of Ecosystem Function. **IEEE Transactions on Geoscience and Remote Sensing**, 57(10), 7740-7751.

[21] **Dashti**, H., Poley, A., Glenn, N.F., Ilangakoon, N., Spaete, L., Roberts, D., Enterkine, J., Flores, A.N., Ustin, S.L., Mitchell, J.J. (2019). Regional Scale Dryland Vegetation Classification with an Integrated Lidar-Hyperspectral Approach. **Remote Sensing**, 11(18), 2141.

[22] Pandit, K., **Dashti**, H., Glenn, N.F., Flores, A.N., Maguire, K.C., Shinneman, D.J., Fellows, A.W. (2019). Developing and optimizing shrub parameters representing sagebrush (Artemisia spp.) ecosystems in the northern Great Basin using the Ecosystem Demography (EDv2.2) model. **Geoscientific Model Development**, 12(11), 4585-4601. https://doi.org/10.5194/gmd-12-4585-2019

[23] Ilangakoon, N.T., Glenn, N.F., **Dashti**, H., Painter, T.H., Mikesell, T.D., Spaete, L.P., Mitchell, J.J., Shannon, K. (2018). Constraining plant functional types in a semi-arid ecosystem with waveform lidar. **Remote Sensing of Environment**, 209, 497-509.

[24] Darvishzadeh, R., Matkan, A.A., **Dashti**, A. (2011). Inversion of a Radiative Transfer Model for Estimation of Rice Canopy Chlorophyll Content. **IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing**, 5(4), 1222-1230.

[25] **Dashti**, A., Darvishzadeh, R., Matkan, A., Hajeb, M. (2012). Inversion of a Radiative Transfer Model for Estimation of Rice Canopy Chlorophyll Content Using Iterative Optimization and ALOS images. **Iranian Journal of Remote Sensing and Geographic Information System**, 12(2). (in Persian)

[26] Shaki, F., Bernard, F., Darvishzadeh, R., **Dashti**, H. (2013). Estimation of Nitrogen Content in Soybean Using Remote Sensing. **Physical Geography Research Quarterly**, 45(2), 102-124. (in Persian)

GRANTS (Funded and Under Reviews)

- **NASA ROSES 2024** — *The Hidden Pulse of Drylands: Global Insights into Ephemeral Vegetation Using PACE Hyperspectral Imaging* (**~$830,000, Funded,** Proposal Lead/Original Co-I)
  Developed concept and wrote proposal; stepped down from Co-I role due to institutional transition 2025
- **Dairy Hub Innovation**-UW Madison — *Assessing the Impact of States and National Climate Policies and Food Behavioral Changes on the US Dairy Industry* (**$150,000, Funded,** Postdoc Fellowship) 2023
- **Northwest Climate Adaptation Science Center**-USDA — *Semi-arid ecosystem characterization using remote sensing and data assimilation* (**~$67,000, Funded**, PhD Fellowship) 2020
- **University of Chicago AI initiative** — *New Frontiers in AI-Driven Knowledge Discovery of Human-Environment Interaction, History, Language, and Material Culture* **(~$250,000, Funded,** Proposal Team Member) 2025

Additional Interdisciplinary Grants:

- University of Chicago Comprehensive Cancer Center — Exploratory *Analysis of Medical Transportation Needs Among Cancer Infusion Patients* (**~$60,000, Funded**, Co-I) 2025

SKILLS

- **Earth System & Ecological Modeling**: Radiative transfer models (PROSAIL, DART), Earth system models (CESM, E3SM), Dynamic Global Vegetation Models (ED2, SiB), Integrated Assessment Models (GCAM), time-series analysis, disturbance detection, and change attribution.
- **Remote Sensing & Environmental Sensing Systems**: Hyperspectral, multispectral, and LiDAR systems (spaceborne, airborne, UAV, terrestrial), GEDI waveform LiDAR, Solar-Induced Fluorescence (OCO-2), terrestrial laser scanning (TLS), UAV-based sensing platforms, plant physiology instruments (LI-COR gas exchange systems, LAI meters, chlorophyll fluorometers, leaf area analyzers), end-to-end field campaign design and deployment (spectrometers, flux towers, TLS, RTK GPS).
- **Programming & Scalable Geospatial Computing**: Python (GDAL, Xarray, Dask, TensorFlow), R, shell scripting; distributed and parallel workflows on HPC/HTC systems; cloud-native geospatial analysis (Microsoft Planetary Computer, STAC, Zarr); reproducible research workflows (Jupyter, Git-based version control).
- **AI & Machine Learning**: Machine learning for regression and classification (Bayesian regression, Random Forest, PLSR, SVM), deep learning for large-scale applications, scalable AI pipelines for Earth observation and ecosystem monitoring.

TEACHING & Mentorship

Instructor

- Special Topics in Remote Sensing and Ecology of the Arctic–Boreal Zone (Graduate/Undergraduate), University of Arizona 2020

Designed & Led Workshops

- Intriduction to Global Change Analysis Model (GCAM), University of Chicago 2025
- Geospatial Python Part 1: Satellite & Climate Raster Analysis, University of Chicago 2025
- Geospatial Python Part 2: Scaling Geospatial Workflows with Dask, University of Chicago 2025
- Python for Analyzing Large Climate and Satellite Data, University of Wisconsin–Madison 2024

Guest Lectures

- *Scientific Programming* (Undergraduate), University of Wisconsin–Madison 2023
- *Advanced Remote Sensing and Image Processing* (Graduate), Boise State University 2018
- *Remote Sensing and Image Processing* (Undergraduate), Boise State University 2017
- *Introduction to Remote Sensing* (Undergraduate), K.N. Toosi University of Technology 2012

Mentorship & Advising

- Mentored **2 PhD students** (radiative transfer modeling, geospatial AI, IAMs), leading to publications.
- Supervised **undergraduate research assistants** in remote sensing and climate data analysis.
- Mentored a **Research Assistant Professor** at UChicago on computational approaches in environmental science.
- Guided **postdocs and early-career researchers** through workshops, hackathons, and collaborative projects.

PROFESSIONAL WORKSHOPS

- PhysFest3 workshop, Kansas State University 2021

- CyVerse Learning - Foundations of Open Science Skills (FOSS), University of Arizona 2020
- Surface Biology and Geology (SBG) Community workshop, DC 2019
- HyspIRI Science and Applications workshop, DC 2018
- FluxCourse workshop, Boulder, CO 2017
- Hyperspectral image and signal processing: Evolution in remote sensing, Los Angeles, CA 2016

PROFESSIONAL and ACADEMIC SERVICES

Reviewer

Remote sensing of the environment; IEEE Transactions on Geoscience and Remote Sensing; Remote sensing; Ecological Applications, Biogeoscience, Journal of Hydrology

Academic Services

- National Ecological Observatory Network, Remote sensing technical advisory group 2021-present
- NASA, Surface Geology Biology, Algorithm development working group 2014-present
- Chair of Scientific Outreach, University of Arizona Postdoctoral Association 2019-2021
- Science Careers for Students, Oscar F. Smith High School 2021
- Conference organizing committee member, "Arizona Postdoctoral Research Conference 2021", 2021
- Preparing underrepresented students for the college entrance exam, Saravan, Iran 2010

CONFERENCES (Including talks and posters)

Dashti, H., Chen, M., Hao, D., Yang, X. (2024). Maximum Light Use Efficiency Shows a Linear Relationship with NIRv and a Non-Linear Relationship with LAI Across Biomes. Ameriflux Annual Meeting 2024.

Dashti, H., Chen, M., Johnson, J.E., Woodgate, W., Vilfan, N., Magney, T., Berry, J.A. (2022). Integrated modeling and observation of the feedback between photosynthesis, leaf optics, active and passive fluorescence, and leaf biochemical and biophysical properties. AGU Fall Meeting, December 2022.

Huo, X., Fox, A.M., **Dashti**, H., Devine, C.J., Raczka, B., Smith, W.K., Anderson, J., Moore, D.J. (2022). Assimilating Leaf Area Index and Aboveground Biomass into CLM Changes Carbon, Water and Energy Cycles in the Arctic and Boreal Region. AGU Fall Meeting, December 2022.

Dashti, H., Smith, W., Huo, X., Fox, A., Devine, C., Moore, D. (2021). Comparing compositional and standard multivariate analyses of land cover data. AGU Fall Meeting, Online, December 2021.

Huo, X., Fox, A., Hoar, T., **Dashti**, H., Devine, C., Smith, W., Anderson, J., Moore, D. (2021). Data assimilation of LAI and biomass into CLM to constrain carbon dynamics in the Arctic and Boreal region. AGU Fall Meeting, Online, December 2021.

Dashti, H., Smith, W.K., Fox, A.M., Devine, C.J., Huo, X., Moore, D.J.P. (2021). The Interconnection Between Land Cover and Land Surface Temperature in the Arctic and Boreal Regions. Ecological Society of America Meeting, Online, August 2021.

Dashti, H., Smith, W.K., Fox, A.M., Devine, C.J., Huo, X., Wang, S., Moore, D.J.P. (2020). The impact of land use land cover change on LST and its drivers in the arctic and boreal regions. AGU Fall Meeting 2020, Online, December 2020.

Biederman, J.A., Smith, W.K., Zhang, F., Pierce, N., Yan, D., Devine, C.J., **Dashti**, H., Roby, M., Scott, R.L., Wang, X., Hu, J., Potts, D.L. (2020). Linking above- and below-ground responses to temporal repackaging of precipitation in a semiarid grassland agroecosystem. AGU Fall Meeting 2020, Online, December 2020.

Enterkine, J., Glenn, N.F., **Dashti**, H., Ustin, S., Knyazikhin, Y. (2020). Canopy structure: the link between optical and lidar remote sensing through canopy spectral invariants. AGU Fall Meeting 2020, Online, December 2020.

Huo, X., Fox, A.M., Hoar, T.J., Anderson, J.L., Smith, W.K., **Dashti**, H., Moore, D.J.P. (2020). Using Data Assimilation of Leaf Area Index to Constrain Decadal Global Carbon Dynamics in the Community Land Model 5.0. AGU Fall Meeting 2020, Online, December 2020.

Dashti, H., Pandit, K., Glenn, N., Shinneman, D., Flerchinge, G., Hudak, A., de Graaff, M., Flores, A., Ustin, S., Ilangakoon, N., Fellows, A. (2019). Evaluating the Performance of a Vegetation Demographic Model (EDv2.2) in Drylands. AGU Fall Meeting 2019, San Francisco, CA, December 2019.

Dashti, H., Glenn, F.N., Illangakoon, N., Ustin, S., De Graaff, M., Flores, L., Qi, Y., Spaete, L. (2017). Mapping vegetation biodiversity of semi-arid ecosystems using hyperspectral remote sensing. ESA 2017, Portland, OR, August 2017.

Dashti, H., Glenn, N.F., Jessica, M., Spaete, L. (2015). Estimation of sagebrush biochemical and biophysical parameters using hyperspectral imagery and inversion of radiative transfer models. 57th Idaho Academy of Science and Engineering (IASE) Annual Meeting and Symposium, Boise, ID, March 2015.

Dashti, H., Glenn, N., Jessica, M., Spaete, L., Lejo, F., Qingtao, Z., Matt, M. (2015). Improving Ecosystem Dynamic Models in a Semi-arid Ecosystem by Integrating Different Sources of Remotely Sensed Data. CSDMS Annual Meeting, Boulder, CO, May 2015.

Dashti, H., Glenn, N., Illangakoon, N., Mitchell, J.J., Dakhal, S., Spaete, L. (2015). Comparison of linear and non-linear regression models to estimate leaf area index of dryland shrubs. AGU Fall Meeting, San Francisco, CA, December 2015.

CHAPTER BOOKS, DATASETS, AND OTHER PRODUCTS

Mitchell, J.J., Glenn, N.F., Dahlin, K.M., Ilangakoon, N.T., **Dashti**, H., Maloney, M.C. (2018). Integrating Hyperspectral and LiDAR Data in the Study of Vegetation. In P.S. Thenkabail, J.G. Lyon, & A. Huete (Eds.), Hyperspectral Remote Sensing of Vegetation (Volume I) (2nd Ed., p. 449). London, New York: CRC Press-Taylor and Francis group.

Dashti, H., Poley, A., Glenn, N., Ilangakoon, N., Spaete, L., Roberts, D., Enterkine, J., Flores, A., Ustin, S., Mitchell, J. (2019). Vegetation maps for Reynolds Creek Experimental Watershed (RCEW) for the year 2015.

Dashti, H., Glenn, N.F., Spaete, L.P., Ilangakoon, N. (2018). Hyperspectral Imagery from AVIRIS-NG for Sites in ID and CA, USA, 2014 and 2015. ORNL Distributed Active Archive Center.

Glenn, N.F., Spaete, L.P., Shrestha, R., Li, A., Ilangakoon, N., Mitchell, J., Ustin, S.L., Qi, Y., **Dashti**, H., Finan, K. (2017). Shrubland Species Cover, Biometric, Carbon and Nitrogen Data, Southern Idaho, 2014. ORNL Distributed Active Archive Center.

PROFESSIONAL ORGINAZATION

- National Ecological Observatory Network, Remote Sensing Technical Group 2023-Now
- American Geophysical Union 2014-Now
- The Ecological Society of America 2017-Now